

Assignment topic: Calculation of Electricity bill of my own household.

Step 1: The number of various equipment available in my household.

- I. Light: 5
- II. Fan: 2
- III. Refrigerator: 1
- IV. TV: 1
- V. Rice cooker: 1

Step 2: The power ratings of every equipment in my household.

- I. Light: 3 lights of 15 watt each and 2 lights of 3 watt each
- II. Fan: 70 watt
- III. Refrigerator: 170 watt
- IV. TV: 100 watt
- V. Rice cooker: 700 watt

Step 3: For each equipment estimate the amount of time used in August.

- I. Light: 3 lights stay on for 7 hours and 2 lights stay on for 2 hours daily.
- II. Fan: Stay on for 15 hours daily.
- III. Refrigerator: Run for 6 hours daily.
- IV. TV: Stay on for 1.5 hours daily
- V. Rice cooker: Used 1.5 hours daily

Step 4: Calculate total energy consumed (in unit)

We know that,

$$\text{Energy} = \frac{\text{power} \times \text{time} \times \text{number of equipment}}{1000}$$

For Light,

$$E_1 = \frac{15 \times 7 \times 3}{1000} \times 31 = 9.7$$

$$E_2 = \frac{3 \times 2 \times 2}{1000} \times 31 = 0.372$$

Total unit for light = $9.7 + 0.372 = 10.072$ u

For Fan,

$$E_1 = \frac{70 \times 15 \times 2}{1000} \times 31 = 65.1u$$

For Refrigerator,

$$E_1 = \frac{170 \times 6 \times 1}{1000} \times 31 = 31.62u$$

For TV,

$$E_1 = \frac{100 \times 1.5 \times 1}{1000} \times 31 = 4.65u$$

For Rice cooker,

$$E_1 = \frac{700 \times 1.5 \times 1}{1000} \times 31 = 32.55u$$

Total energy consumed = $(10.072 + 65.1 + 31.62 + 4.65 + 32.55) = 143.992u$

Step 5: Total cost of unit and other costs-

Total cost of unit = $(143.992 \times 7) = 1007.94$ taka

Meter rent = 40 taka

Demand charge = 84 taka

Vat_(5%) = $(1007.94 \times 5\%) = 50.4$ taka

Step 6: Total electricity bill-

Total electricity bill of august = $(1007.94 + 40 + 84 + 50.4) = 1182.34$ taka

Step 7: Compare this bill with the bill of paper-

Electricity bill of august on paper of bill = 1327.45 taka

Compare = $(1327.45 - 1182.34) = 145.11$ taka

Step 8: The reason for variation-

1. **Estimation Errors:** The estimated usage times or power ratings might not match the actual usage.

2. **Standby Power Consumption:** Many devices consume power even when turned off, which might not have been included.
3. **Tariff Variations:** The actual billing might include additional charges or different rates that weren't considered in the calculation.
4. **Meter Accuracy:** Differences between the household's meter readings and the utility company's recorded consumption could cause discrepancies.